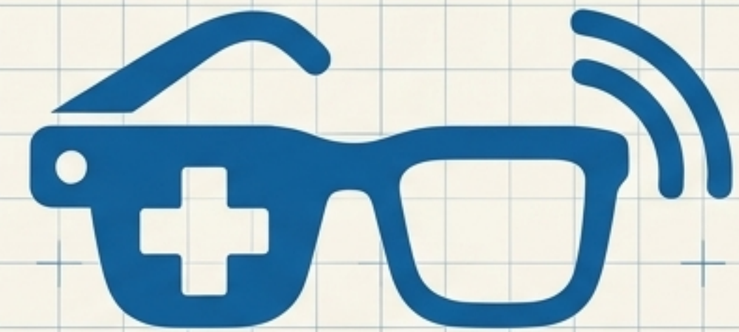


MedGlass — Fully-Local Ambient Clinical Scribe

Five AI Models.

One AMD Radeon GPU.

Zero PHI Leaves the Room.

AMD AI DevMaster Hackathon 2026 · Track 1: Multimodal Content Creation Tools



The Need: AI Scribes

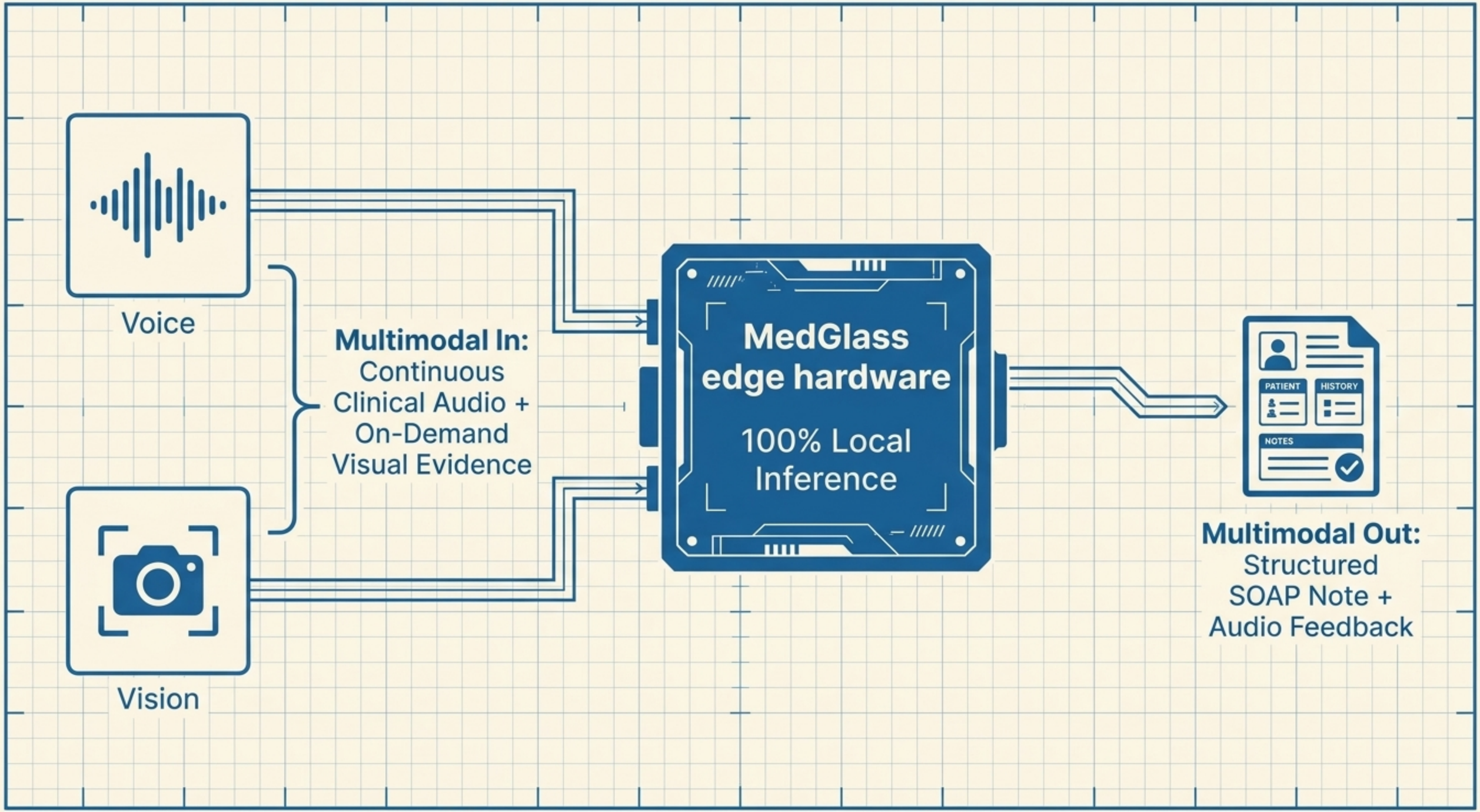
Clinicians require ambient voice and visual assistance to reduce severe documentation burden.



The Constraint: PHI Security

Processing sensitive clinical conversations via traditional cloud APIs creates massive, unacceptable HIPAA/privacy risks.

The Solution requires local intelligence. We must run a state-of-the-art multimodal pipeline entirely at the edge.


Voice
Vision
Multimodal In:
Continuous
Clinical Audio +
On-Demand
Visual Evidence
MedGlass
edge hardware
100% Local
Inference
PATIENT
HISTORY
NOTES
Multimodal Out:
Structured
SOAP Note +
Audio Feedback

Session Start



Session Start

Voice announces patient. Rolling 5s transcription begins.

Live Capture



Live Capture

Live transcript generated in real-time while clinician speaks.

Visual Evidence



Visual Evidence

Add photo via glasses. Screen flicker confirms capture. (× to delete).

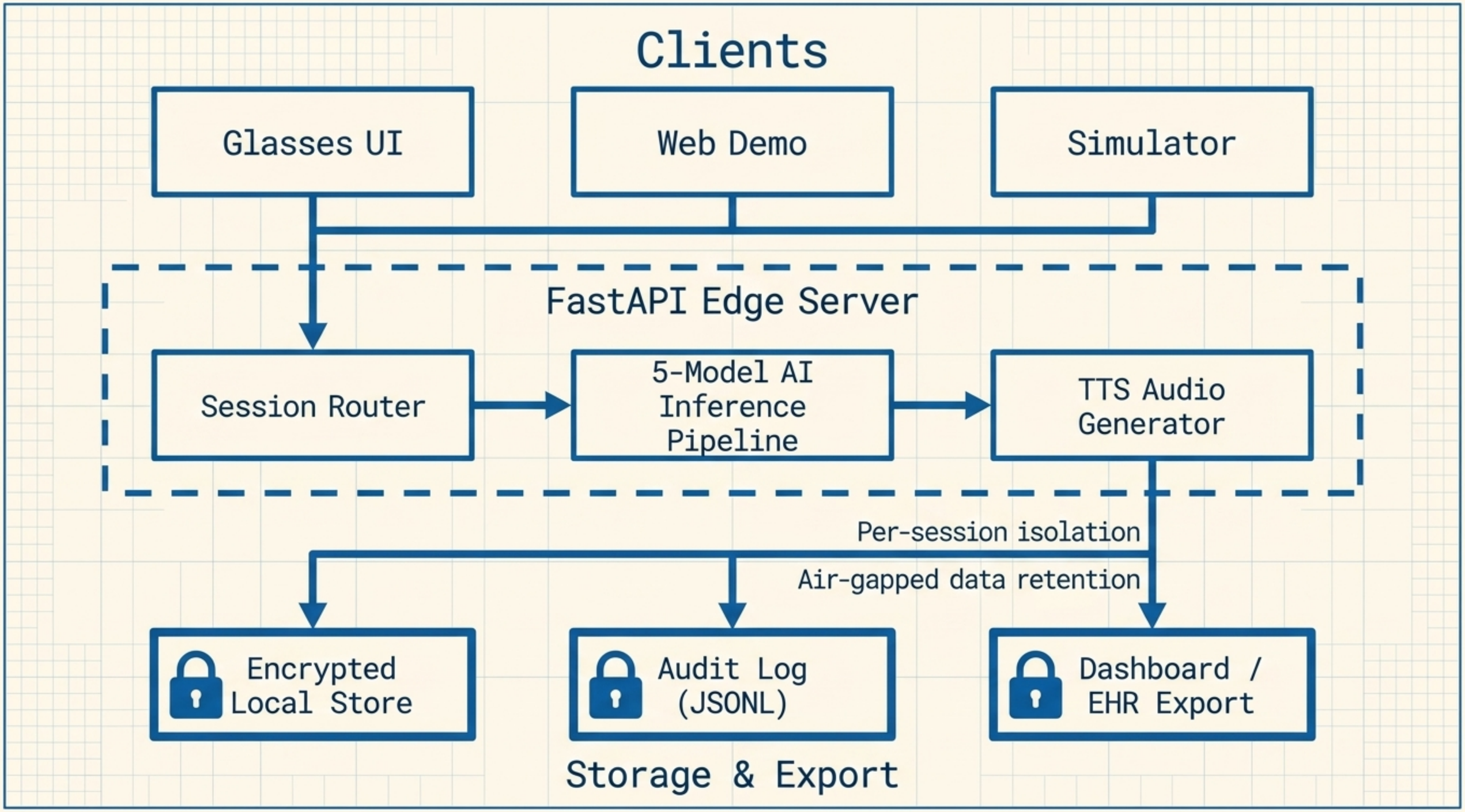
Secure Save



Secure Save

Stop session. Note saved instantly with transcript + photo record.

Dashboard Callout: Web Dashboard features S/O/A/P drawer layout, EHR downloads, Check-off next steps.


Clients
Glasses UI
Web Demo
Simulator
FastAPI Edge Server
Session Router
5-Model AI Inference Pipeline
TTS Audio Generator
Per-session isolation
Air-gapped data retention
Encrypted Local Store
Audit Log (JSONL)
Dashboard / EHR Export
Storage & Export

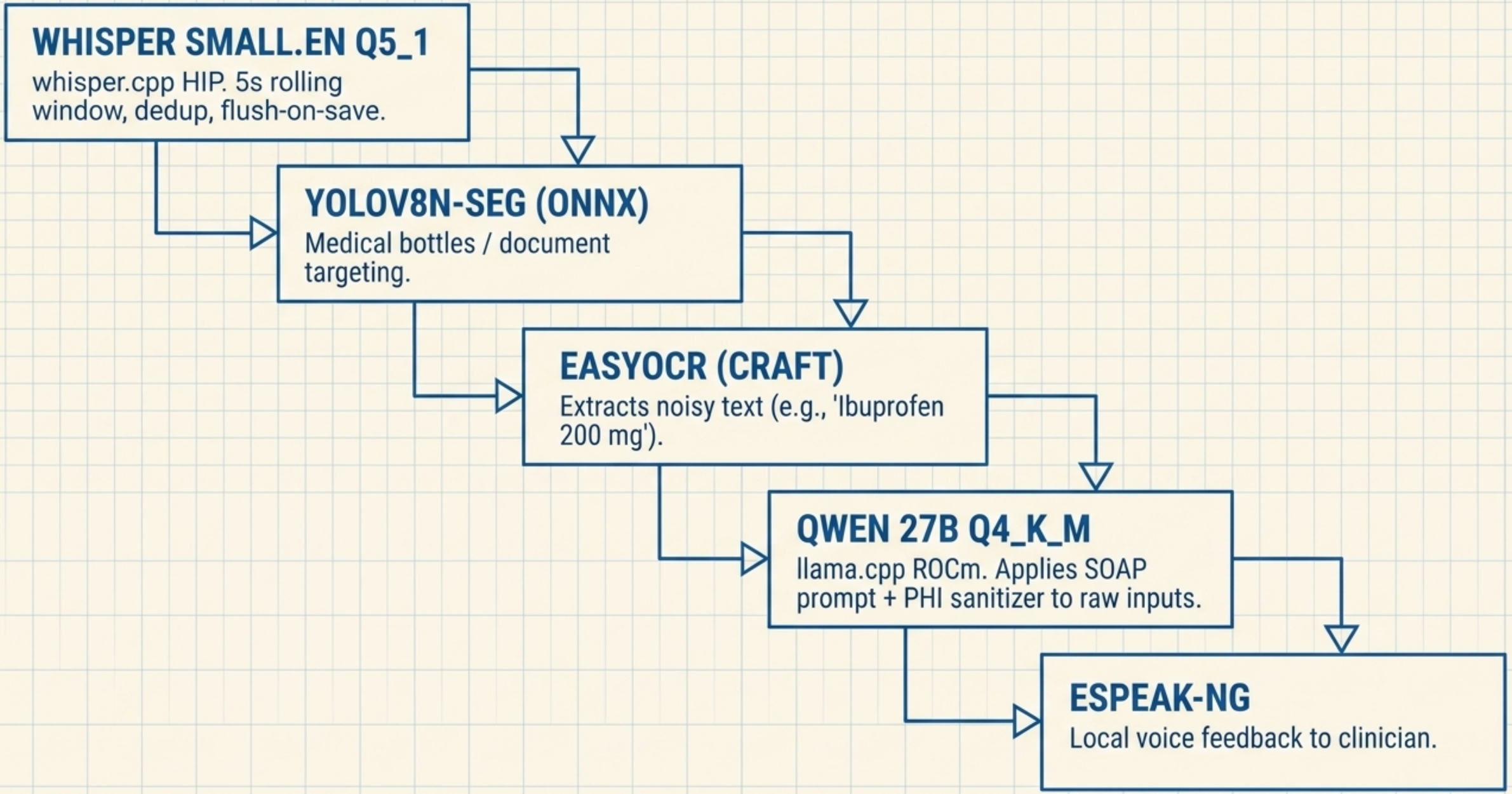

WHISPER SMALL.EN Q5_1
whisper.cpp HIP. 5s rolling window, dedup, flush-on-save.
YOLOV8N-SEG (ONNX)
Medical bottles / document targeting.
EASYOCR (CRAFT)
Extracts noisy text (e.g., 'Ibuprofen 200 mg').
QWEN 27B Q4_K_M
llama.cpp ROCm. Applies SOAP prompt + PHI sanitizer to raw inputs.
ESPEAK-NG
Local voice feedback to clinician.

Exhibit A: Glasses-Cam OCR Integration

Glasses POV → medication OCR (EasyOCR)

640×480 glasses cam feed · photo saved · OCR on captured JPEG



Detected: Ibuprofen 200 mg tablets (NSAID)

Pipeline: EasyOCR on photo_stere JPEG · meds matched to Rx vocabulary

Raw OCR (noisy, brand misread):

```
"rofel Tablets,_ VPain Reliever/ Fever Reducer (NSAID) ... Ibu;' 200mg Coa
```

Normalized finding:

```
"Ibuprofen 200 mg tablets (NSAID) – brand OCR errors present"
```

Appears in note Objective section + photo record:

/photos/default_1785956467_273d.jpg

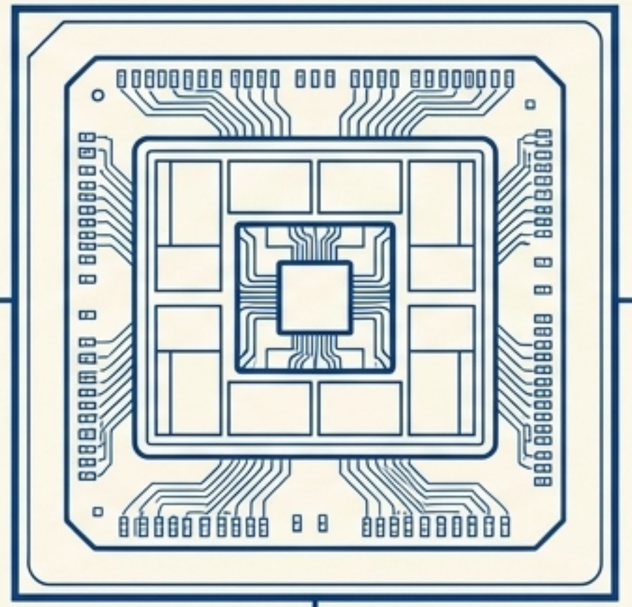
OCR stage: CPU ok · GPU ok · 2.5 s per frame

Overcoming Noise: Pipeline successfully normalizes noisy brand misreads (e.g., 'rofel Tablets') into structured clinical data.

Clinical Output: 'Ibuprofen 200 mg tablets (NSAID)' accurately extracted at 2.5s per frame and pushed to EHR Objective section.

Engine Adaptation: Radeon AI PRO R9700

Architecture: RDNA4 / Navi 48 hardware.



Stack: AMD ROCm 7.1 Integration.

Execution: Simultaneous multimodal execution (whisper.cpp + llama.cpp on ROCm) without memory collision

The primary technical achievement is balancing 5 distinct AI models—including a massive 27B parameter LLM—concurrently on a single edge GPU.

Exhibit B: GPU Benchmark Data

6.3× Speedup

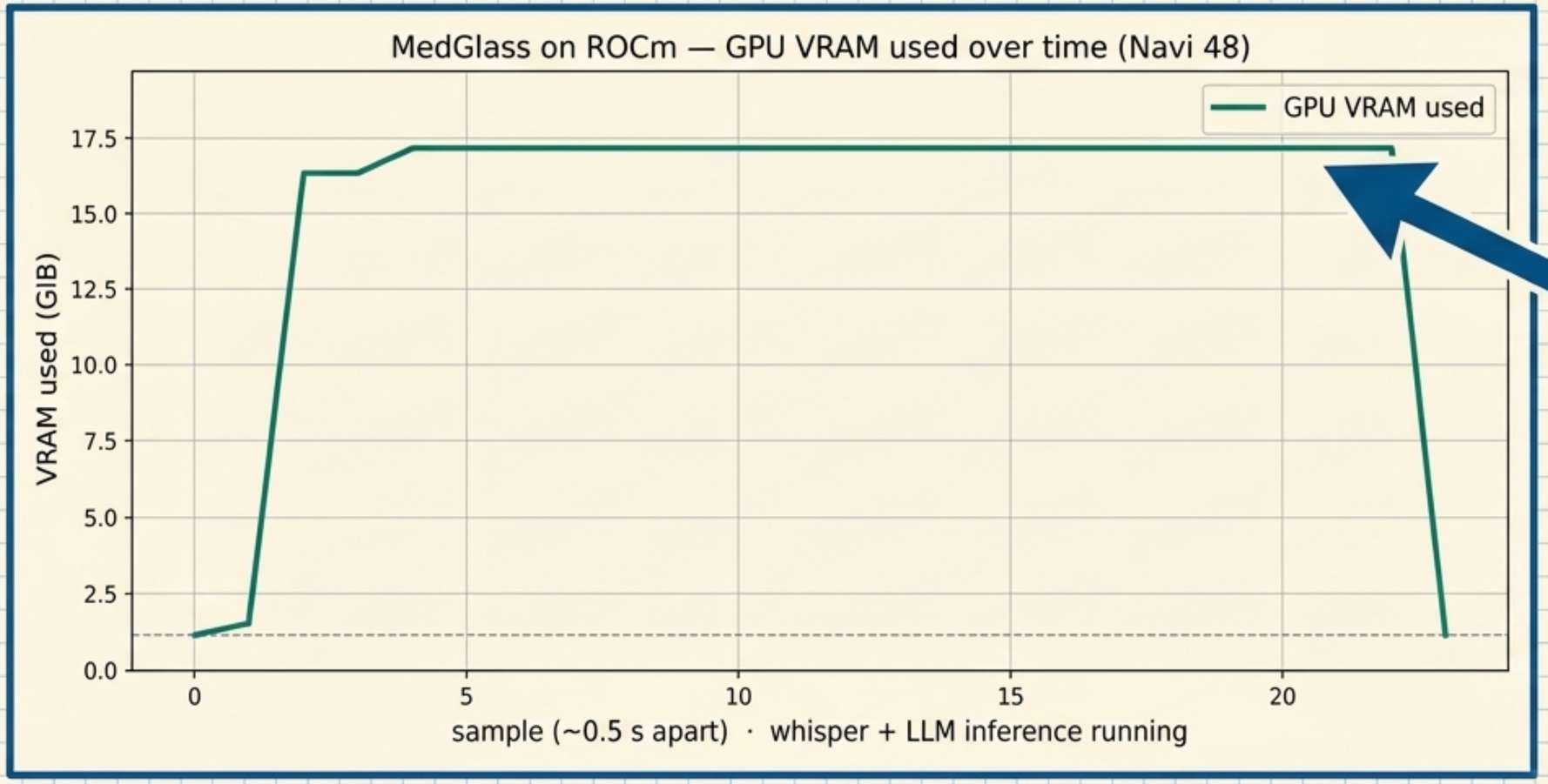
Whisper STT inference reduced from 6.96s (CPU) to a near-instant 1.10s via ROCm.

Stage	CPU	GPU (ROCm)	Speedup
whisper.cpp ggml-small.en	6.96 s	1.10 s	6.3×
LLM Qwen3-27B Q4_K_M (all 999 layers)	25.68 s	7.23 s	3.6×
Vision YOLOv8n-seg	88 ms/ frame	unavailable*	—
OCR EasyOCR	ok	ok	—

3.6× Speedup

Qwen3-27B LLM (all 999 layers) execution reduced from 25.68s to 7.23s, enabling fluid real-time clinical summarization.

Sustaining a 27B Model on the Edge.

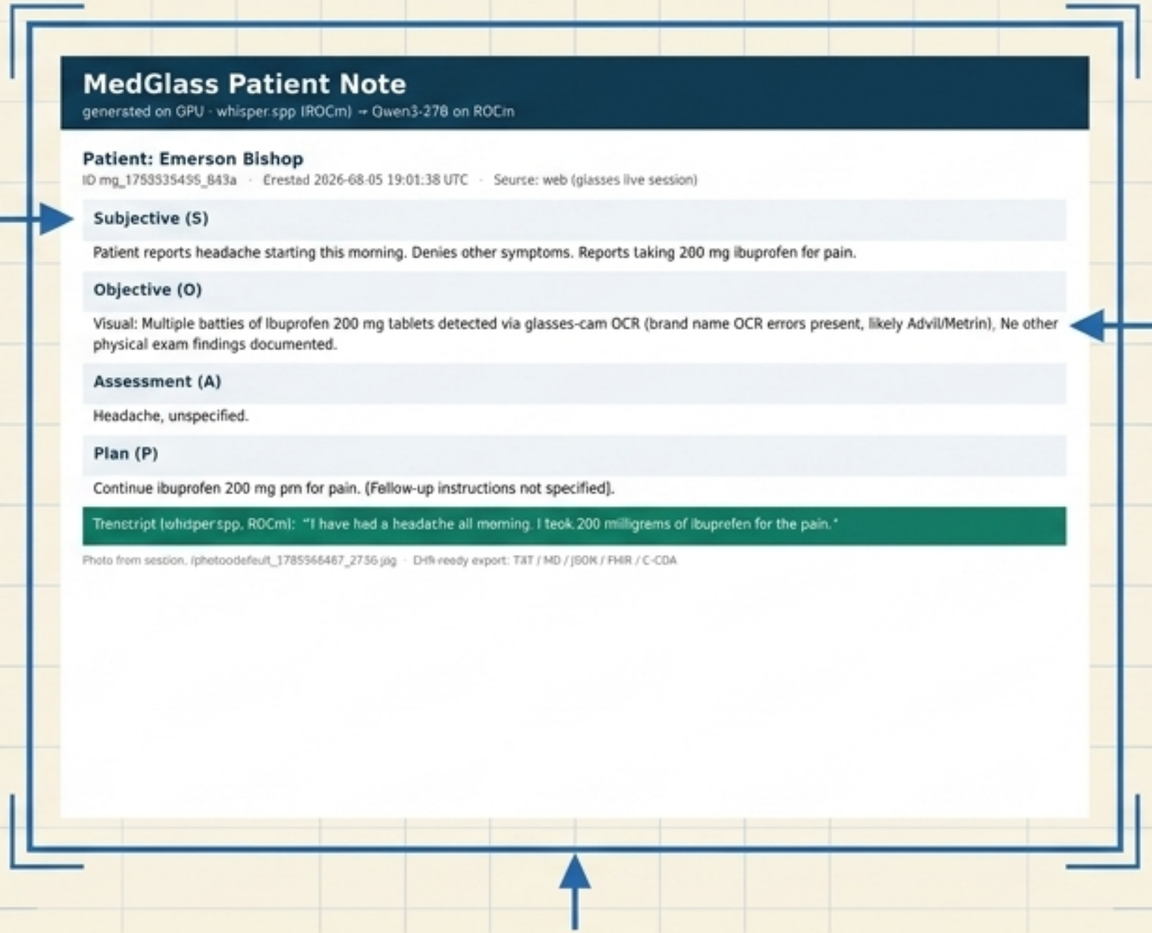


17.2 GiB Peak VRAM Utilization.

GPU Proof: whisper + LLM inference running concurrently. rocm-smi confirms stable execution strictly within the limits of the Radeon hardware, effectively air-gapping the system.

Exhibit D: EHR-Ready Note Generation

Audio Transcript -> Automatically structured into Subjective narrative.



OCR Pipeline -> Injects detected medication (Ibuprofen 200mg) directly into the physical exam record.

Export Ready -> Native support for TXT, MD, JSON, FHIR, and C-CDA hand-offs.

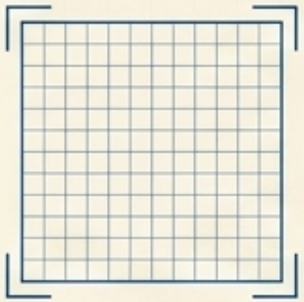
The Zero-PHI Guarantee (docs/HIPAA_COMPLIANCE.md)

Security Protocol	Clinical Compliance Result
Scrypt password hashing & TOTP 2FA	Ensures strict local access control.
Encryption at rest & JSONL audit log	Mandatory HIPAA traceability and data security.
No-store caching for PHI & Retention purge	Zero residual patient data leaks post-session.
CSRF double-submit & IP allow-list	Network hardening against local intrusion.

By moving the 27B model to the edge, the network attack surface is virtually eliminated.

The Conversation Never Leaves the Room.

[1 AMD R9700 GPU] + [5 Open-Source AI Models] = [Infinite Local Scaling]



QR Code

Project Repository: README.md, Architecture Docs, and Setup.

Live Demonstration: /demo/ directory and recorded session.